SAIS: Semantic Anchor & Intent Score

A Fact-Aware, Structure-Aware Metric for Reference-Based Evaluation of Generated Text

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Abstract

Automatic metrics for generated text often prioritize lexical overlap or semantic similarity, leaving factual completeness, unsupported additions, ordering, and explicit requirements only indirectly represented. SAIS combines normalized embedding similarity, idea-block alignment, factual precision and recall, explicit constraints, and symmetric length regularization. On 19,725 valid WMT22 English-to-German expert-MQM segments, a local Qwen embedding configuration produced stronger system-level correlation than BLEU and ROUGE-L, but weaker segment-level correlation. The result is evidence of limited system-level promise, not overall superiority.

1. Motivation

BLEU and ROUGE are useful overlap metrics, but they can under-credit valid paraphrase and do not explicitly verify required facts or hard requirements. Embedding metrics better represent semantic similarity but still do not ensure that a candidate preserved dates, numbers, entities, causal order, or safety constraints. SAIS is an interpretable reference-based metric intended to expose these separate dimensions.

2. Formulation

For candidate C and reference R:

$$\mathrm{SAIS}(C,R) = [\alpha\, S_{\mathrm{mesh}}(C,R) + (1-\alpha)\, S_{\mathrm{logic}}(C,R)] \times S_{\mathrm{fact}}(C,R) \times \Gamma(C,R) \times \mathrm{BP}(C,R)$$

Meaning Mesh

Document embeddings u and v are compared with normalized cosine similarity: $S_{\mathrm{mesh}} = (1 + \cos(u,v)) / 2$. This maps the cosine range to [0,1].

Logic Flow

Candidate and reference are segmented into ordered idea blocks. Each reference block aligns to its best candidate block, weighted by normalized positional distance: $\exp(-\lambda\, |i/m - j/n|)$. This avoids assuming equal numbers of blocks.

Fact Anchor

Required and candidate facts are canonicalized and matched one-to-one. Dates, numbers, aliases, capitalization, and whitespace can be normalized. Precision penalizes candidate-only facts; recall penalizes missing reference facts; their Fβ combination forms S_{fact}.

Constraint Gate and Length Penalty

Explicit required or forbidden conditions receive failure factors from 0 to 1. A hard failure uses zero. The symmetric length penalty is exp(−β |log(|C| / max(|R|,1))|), discouraging both truncation and padding.

3. Implementation

The Python implementation returns a final score and diagnostics: component scores, matched facts, missing facts, candidate-only facts, and violated constraints. It supports sentence-transformer embeddings, a local Ollama embedding adapter, and a deterministic hashing encoder for non-semantic tests. The empirical run used the local qwen3-embedding:0.6b Ollama model.

4. WMT22 Expert MQM Evaluation

Setup

The evaluation used the WMT22 English-to-German expert MQM release. Rows without a usable hypothesis, reference, or human score were excluded, leaving 19,725 segments across 15 systems. Raw MQM values are negative error penalties, so larger values, closer to zero, are better. SAIS used default parameters α=0.6, λ=3.0, length β=0.2, fact β=1.0, automatic fact extraction, and no curated constraints.

Metric	Segment Pearson	Segment Spearman	System Pearson	System Spearman
SAIS, Qwen3 embedding	0.075	0.189	**0.727**	**0.539**
BLEU	0.199	0.228	0.632	0.457
ROUGE-L	**0.227**	**0.258**	0.712	0.500

Table 1. Correlation with expert human MQM quality judgments.

Interpretation

SAIS has the strongest observed system-level correlations, though its margin over ROUGE-L is small and rests on only 15 systems. At segment level, both lexical baselines correlate more strongly with human MQM. This experiment does not establish SAIS as a generally superior translation metric.

5. Limitations

Translation does not fully exercise SAIS's intended factual and explicit-constraint components. The evaluation uses automatic rather than curated facts, one language direction, no BERTScore baseline, and no confidence intervals or significance tests. SAIS inherits embedding bias and can be affected by segmentation, entity extraction, incomplete references, and alias errors. It must not be the sole basis for medical, legal, employment, educational, safety, or other high-impact decisions.

6. Conclusion

SAIS separates semantic similarity, logical order, factual precision and recall, explicit constraints, and length regularization in one interpretable metric. Its multiplicative fact and constraint terms prevent high semantic similarity from hiding certain critical failures. The WMT22 result is mixed: system-level alignment is promising while segment-level alignment is not competitive with the lexical baselines tested. Further calibration and validation on fact-sensitive tasks are required.

References

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